medRxiv preprint doi: https://doi.org/10.1101/2021.05.12.21257083; this version posted September 17, 2021. The copyright holder for this preprint **(which was not certified by peer review)** is the author/funder, who has granted medRxiv a license to display the preprint in perpetuity.
It is made available under a CC-BY 4.0 International license .

Background rates of five thrombosis with thrombocytopenia syndromes of special interest for COVID-19 vaccine safety surveillance: incidence between 2017 and 2019 and patient profiles from 25.4 million people in six European countries

Edward Burn[1,2], Xintong Li[2], Kristin Kostka[3,4], Henry Morgan Stewart[3], Christian Reich[3], Sarah Seager[3], Talita Duarte-Salles[1], Sergio Fernandez-Bertolin[1], María Aragón[1], Carlen Reyes[1], Eugenia Martinez-Hernandez[5], Edelmira Marti[6], Antonella Delmestri[2], Katia Verhamme[7], Peter Rijnbeek[7], Scott Horban[8], Daniel R Morales[8], Daniel Prieto-Alhambra[2,7]*

[1]Fundació Institut Universitari per a la recerca a l'Atenció Primària de Salut Jordi Gol i Gurina (IDIAPJGol), Barcelona, Spain, [2]Centre for Statistics in Medicine (CSM), Nuffield Department of Orthopaedics, Rheumatology and Musculoskeletal Sciences (NDROMS), University of Oxford, UK, [3]Real World Solutions, IQVIA, Cambridge, MA, US, [4]The OHDSI Center at The Roux Institute, Northeastern University, Portland, ME, US,[5]Department of Neurology, Hospital Clinic and University of Barcelona, Barcelona, Spain, [6]Hemostasis and Thrombosis Unit, Hematology Department, Hospital Clínico Universitario de Valencia, Spain, [7]Department of Medical Informatics, Erasmus University Medical Center, Rotterdam, The Netherlands, [8]Division of Population Health and Genomics, University of Dundee, Dundee, UK

*corresponding author

Prof Daniel Prieto-Alhambra, Botnar Research Centre, Windmill Road, OX37LD, Oxford, UK, daniel.prietoalhambra@ndorms.ox.ac.uk

NOTE: This preprint reports new research that has not been certified by peer review and should not be used to guide clinical practice.

medRxiv preprint doi: https://doi.org/10.1101/2021.05.12.21257083; this version posted September 17, 2021. The copyright holder for this preprint (which was not certified by peer review) is the author/funder, who has granted medRxiv a license to display the preprint in perpetuity.
It is made available under a CC-BY 4.0 International license .

Abstract

Background

Thrombosis with thrombocytopenia syndrome (TTS) has been reported among individuals vaccinated with adenovirus-vectored COVID-19 vaccines. In this study we describe the background incidence of TTS in 6 European countries.

Methods

Electronic medical records from France, Netherlands, Italy, Germany, Spain, and the United Kingdom informed the study. Incidence rates of cerebral venous sinus thrombosis (CVST), splanchnic vein thrombosis (SVT), deep vein thrombosis (DVT), pulmonary embolism (PE), and stroke, all with concurrent thrombocytopenia, were estimated among the general population between 2017 to 2019. A range of additional adverse events of special interest for COVID-19 vaccinations were also studied in a similar manner.

Findings

A total of 25,432,658 individuals were included. Background rates ranged from 1.0 (0.7 to 1.4) to 8.5 (7.4 to 9.9) per 100,000 person-years for DVT with thrombocytopenia, from 0.5 (0.3 to 0.6) to 20.8 (18.9 to 22.8) for PE with thrombocytopenia, from 0.1 (0.0 to 0.1) to 2.5 (2.2 to 2.7) for SVT with thrombocytopenia, and from 0.2 (0.0 to 0.4) to 30.9 (28.6 to 33.3) for stroke with thrombocytopenia. CVST with thrombocytopenia was only identified in one database, with incidence rate of 0.1 (0.1 to 0.2) per 100,000 person-years. The incidence of TTS increased with age, with those affected typically having more comorbidities and greater medication use than the general population. TTS was also more often seen in men than women. A sizeable proportion of those affected were seen to have been taking antithrombotic and anticoagulant therapies prior to their TTS event.

Interpretation

Although rates vary across databases, TTS has consistently been seen to be a very rare event among the general population. While still very rare, rates of TTS are typically higher among older individuals, and those affected were also seen to generally be male and have more comorbidities and greater medication use than the general population.

Funding

medRxiv preprint doi: https://doi.org/10.1101/2021.05.12.21257083; this version posted September 17, 2021. The copyright holder for this preprint **(which was not certified by peer review)** is the author/funder, who has granted medRxiv a license to display the preprint in perpetuity.
It is made available under a CC-BY 4.0 International license .

This study was funded by the European Medicines Agency (EMA/2017/09/PE Lot 3).

medRxiv preprint doi: https://doi.org/10.1101/2021.05.12.21257083; this version posted September 17, 2021. The copyright holder for this preprint (which was not certified by peer review) is the author/funder, who has granted medRxiv a license to display the preprint in perpetuity.
It is made available under a CC-BY 4.0 International license .

Introduction

In little over a year since the beginning of the coronavirus disease 2019 (COVID-19) pandemic, numerous vaccines against SARS-CoV-2 have been developed based on several platforms.[1] A range of these have demonstrated a high degree of efficacy in large phase 3 clinical trials,[2–4] received conditional approvals from regulators, and together they have already been given to over a billion individuals.[5] The benefits of these vaccines are demonstrable with, for example, a large study on mass vaccination in Israel finding the estimated effectiveness of BNT162b2 mRNA vaccine to be 94% for symptomatic COVID-19, 87% for hospitalisation, and 92% for severe COVID-19 from seven days after the second dose.[6] Similarly, the use of the BNT162b2 mRNA and ChAdOx1 in Scotland have been associated with substantial reductions in the risk of developing severe COVID-19 disease.[7]

There remains, however, a need to continue assessing the safety of vaccines against SARS-CoV-2 and assess safety signals as and when they arise. While the results of phase 3 clinical trials provided valuable information on the rates of relatively common, but mostly mild, adverse reactions following vaccination against SARS-CoV-2, they were not powered to study the occurrence of rare adverse events of special interest. While the risks of rare but serious adverse events might be low, nationwide vaccination campaigns where millions of people are inoculated can lead to a considerable absolute number of any such events to occur.

A particular area of concern has recently arisen relating to the occurrence of thrombosis (often cerebral or abdominal) with concomitant thrombocytopenia among individuals who had received adenovirus-based vaccine against SARS-CoV-2. As of the 28th April 2021, 242 instances of thromboembolic events with thrombocytopenia among individuals who had recently received the ChAdOx1 vaccine in the UK had been identified on the basis of spontaneous reports. Of these, cerebral venous sinus thrombosis (CVST) was reported in 93 of the cases.[8] Meanwhile, as of the 23rd April 2021, 15 confirmed reports of thrombosis with thrombocytopenia syndrome (TTS) had been identified for the Ad.26.COV2.S vaccine in the US.[9] These spontaneous reports of TTS among individuals who had received the ChAdOx1 and Ad.26.COV2.S vaccines came at a time when 22.6 million first doses and 5.9 million second doses of the ChAdOx1 vaccine had been given in the UK and more than 8 million doses of the Ad.26.COV2.S had been given in the US.[8,9]

The degree to which the reported TTS events exceed the number that would typically be observed in the absence of vaccinations against SARS-CoV-2 is not yet well-known, nor is how the profiles of those with such events after vaccination have differed from those who typically experience them. Establishing the rates of TTS events among the general population in previous years will help to

medRxiv preprint doi: https://doi.org/10.1101/2021.05.12.21257083; this version posted September 17, 2021. The copyright holder for this preprint (which was not certified by peer review) is the author/funder, who has granted medRxiv a license to display the preprint in perpetuity.
It is made available under a CC-BY 4.0 International license .

provide context for the observed rates being seen among those vaccinated.[10] Moreover, a description of the profiles of the individuals who have had TTS events in the past will also help to inform a consideration of whether the profiles of individuals with TTS after a vaccination against COVID-19 differ to those who typically have such events.

In this study we set out to estimate the background incidence rates of TTS and to describe the profiles of individuals who typically have these types of events. We did this using electronic medical records collected between 2017 to 2019 and covering over 25 million people across six European countries. In addition, we performed similar analyses for a range of other embolic and thrombotic events and coagulopathies of special interest for COVID-19 vaccinations.

Methods

Study Design

An international network cohort study using routinely-collected primary care data from across Europe. Data were previously mapped to the Observational Medical Outcomes Partnership (OMOP) Common Data Model (CDM), which allowed for the study to be run in a distributed manner, with common analytic code run by each site without the need to share patient-level data between sites.[11–13]

Data sources

Data from seven electronic medical records databases from France, Netherlands, Italy, Germany, Spain, and the United Kingdom informed the analysis. The Clinical Practice Research Datalink (CPRD) GOLD database contains data contributed by general practitioners (GP) from the United Kingdom.[14] The Health Informatics Centre at the University of Dundee (HIC Dundee) database includes linked primary care and hospital data for persons from the Tayside region of Scotland, capturing around 20% of the Scottish population. The Integrated Primary Care Information (IPCI) database is collected from electronic healthcare records of patients registered with GPs throughout the Netherlands. IQVIA Longitudinal Patient Data (LPD) Italy includes anonymised patient records collected from software used by GPs during an office visit to document patients´ clinical records. IQVIA LPD France is a computerised network of physicians including GPs who contribute to a centralised database of anonymised patient electronic medical records.[15] IQVIA Disease Analyser (DA) Germany is collected

medRxiv preprint doi: https://doi.org/10.1101/2021.05.12.21257083; this version posted September 17, 2021. The copyright holder for this preprint (which was not certified by peer review) is the author/funder, who has granted medRxiv a license to display the preprint in perpetuity.
It is made available under a CC-BY 4.0 International license .

from extracts of patient management software used by GPs and specialists practicing in ambulatory care settings. The Information System for Research in Primary Care (SIDIAP; www.sidiap.org) is a primary care records database that covers approximately 80% of the population of Catalonia, North-East Spain. SIDIAP can be linked to the minimum basic set of hospital discharge data (CMBD- HA), which includes diagnosis and procedures registered during hospital admissions.[16] Results for SIDIAP CMBD-HA are presented in this manuscript, with results for SIDIAP only also reported in the supplementary materials for comparison.

In summary, all the included databases captured outpatient diagnoses and lab measurements. SIDIAP CMBDH-HA and HIC Dundee also directly captured diagnoses from linked hospital data. HIC Dundee was the only database that, in addition, included hospital lab measurements. Study participants and time at risk

The primary study cohort consisted of individuals present in a database as of the 1st January 2017, with this date used as the index date for all study participants. These individuals were followed up to whichever came first: the outcome of interest, exit from the database, or the 31st December 2019 (the end of study period). A second study cohort definition of active patients was considered, where individuals entered the cohort on the date of their first visit occurrence after 1st January 2017. As with the primary study cohorts these individuals were followed up to whichever came first: the outcome of interest, exit from the database, or 31st December 2019. As a sensitivity analysis, study cohorts were also generated with the additional requirement that individuals had a minimum of one year of history available in the database prior to their index date. For each analysis of a given outcome, any individual with that outcome in the year prior to their index date was excluded.

Outcomes

Here we summarise results for five specific TTS events of interest: CVST with thrombocytopenia, DVT with thrombocytopenia, PE with thrombocytopenia, SVT with thrombocytopenia, and stroke with thrombocytopenia. Occurrences of CVST, DVT, PE, and stroke were identified on the basis of diagnostic codes. Given the limited granularity in coding for a number of the databases, stroke encapsulated both ischemic and hemorrhagic stroke. Thrombocytopenia was identified either by a diagnostic code or a measurement of between 10,000 to 150,000 platelets per microliter of blood and was required to have been observed over a time window starting ten days prior to the event of interest up to ten days afterwards. For comparison, we also provide results for each of the outcomes without the need for them to be accompanied by thrombocytopenia. In addition, we provide background rates for coagulopathies that have been identified as potential causes for TTS: heparin-

medRxiv preprint doi: https://doi.org/10.1101/2021.05.12.21257083; this version posted September 17, 2021. The copyright holder for this preprint (which was not certified by peer review) is the author/funder, who has granted medRxiv a license to display the preprint in perpetuity.
It is made available under a CC-BY 4.0 International license .

induced thrombocytopenia (HIT), disseminated intravascular coagulation (DIC), immune thrombocytopenia, and thrombotic thrombocytopenic purpura (TTP - which included hemolytic uremic syndrome).

The outcomes described here are taken from a wider set of adverse events of special interest (AESI) for COVID-19 vaccinations. Three sets of outcome events were identified: 1) venous thromboembolic events, 2) arterial thromboembolic events, and 3) rare embolic, coagulopathies, and TTS events. For venous thromboembolic events, instances of DVT (with one broad definition and another narrow - in this manuscript, we describe results for the narrow definition) and PE events were identified, with venous thromboembolism events defined as the occurrence of either DVT or PE. For arterial thromboembolic events, instances of myocardial infarction and ischemic stroke were identified, along with the composite outcome of either of these events. Instances of stroke, either ischemic or hemorrhagic, were also identified. A wide set of rare embolic and thrombotic events and thrombocytopenias and platelet disorders were considered: DIC, immune thrombocytopenia, TTP, HIT, thrombocytopenia, thrombotic thrombocytopenic purpura, platelet disorder/s, CVST, splenic vein thrombosis, splenic artery thrombosis, splenic infarction, hepatic vein thrombosis, hepatic artery thrombosis, portal vein thrombosis, intestinal infarction, mesenteric vein thrombosis, celiac artery thrombosis, visceral vein thrombosis, and SVT.

All study outcomes were identified using code lists reviewed by three epidemiologists (EB, DPA, CR), a clinical neurologist (EMH) and a clinical haematologist (EM). These definitions were reviewed using the aid of the CohortDiagnostics R package,[17] so as to identify additional codes of interest and to remove those highlighted as irrelevant based on feedback from regulators (e.g. puerperium and pregnancy-related disease) through an iterative process during the initial stages of analyses. A detailed description of the definitions used to identify the outcomes of the study is provided at https://livedataoxford.shinyapps.io/CovCoagOutcomesCohorts/. This application summarises the codes used to identify outcomes and their frequency in the databases used in the study, the overlap between cohorts in the databases as a whole, and a detailed summary of the profiles of all the individuals with a code of interest in each of the databases.

Patient profiles

The characteristics of the study population were extracted relative to their index date, as were those of individuals with a particular outcome of interest relative to the date of their event. The age and sex of individuals was identified, along with their history of conditions and medication use. Using all of an individual's prior observation time, prior diagnosis of autoimmune disease, antiphospholipid

medRxiv preprint doi: https://doi.org/10.1101/2021.05.12.21257083; this version posted September 17, 2021. The copyright holder for this preprint (which was not certified by peer review) is the author/funder, who has granted medRxiv a license to display the preprint in perpetuity.
It is made available under a CC-BY 4.0 International license .

syndrome, thrombophilia, asthma, atrial fibrillation, malignant neoplastic disease, diabetes mellitus, obesity, heart disease, hypertensive disorder, renal impairment, chronic obstructive pulmonary disease (COPD), or dementia were identified. Meanwhile, prior medication use was identified using a time window of 183 days prior to four days prior the index date. Drug exposures of antithrombotic and anticoagulant therapies, non-steroidal anti-inflammatory drugs, Cox-2 inhibitors, systemic corticosteroids, lipid modifying agents, antineoplastic and immunomodulating agents, hormonal contraceptives for systemic use, tamoxifen, and sex hormones and modulators of the genital system were identified over this time window.

Statistical methods

The profiles of the study cohorts and those with an outcome of interest were summarised, with median and interquartile range (IQR) used for continuous variables and counts and percentages used for categorical variables. For each study outcome, the number of events, the observed time at risk, and the incidence rate per 100,000 person-years are summarised along with 95% confidence intervals. These results are provided for the study cohorts and stratified by data source as a whole and stratified by age (≤44, 45 to 64, or ≥65 years old) and sex. To aid in comparison with rates being reported after vaccinations, the expected number of events per 28 days for a population of 10 million were calculated based on the incidence rates calculated for the overall study cohorts and age strata.

Code availability

All analytic code used for the analysis has been made publicly available at https://github.com/oxford-pharmacoepi/CovCoagBackgroundIncidence.

Role of funding source

This study was funded by the European Medicines Agency (EMA). The study outcomes were chosen in collaboration with the EMA so as to best reflect the safety signals of interest. The study protocol was reviewed by the EMA prior to its submission, and registered in the European Union electronic Register of Post-Authorisation Studies (EU PAS Register®): http://www.encepp.eu/encepp/viewResource.htm?id=40415

medRxiv preprint doi: https://doi.org/10.1101/2021.05.12.21257083; this version posted September 17, 2021. The copyright holder for this preprint (which was not certified by peer review) is the author/funder, who has granted medRxiv a license to display the preprint in perpetuity.
It is made available under a CC-BY 4.0 International license .

Results

A total of 25,432,658individuals were included in the study (3,913,071 from CPRD, 948,561 from HIC Dundee, 8,459,098 DA Germany, 3,951,633 LPD France, 1,299,288 IPCI, 1,066,230 LPD Italy, and 5,794,777 from SIDIAP CMBDH-HA). The median age of the background populations ranged from 41 in CPRD to 52 in DA Germany and LPD Italy. More detailed characteristics of each of these populations are summarised in Table 1.

Incidence rates for the different outcomes of thrombosis and TTS across the databases are summarised in Table 2. The incidence rates for CVST ranged from 0.3 (95% CI: 0.2 to 0.5) to 1.2 (1.0 to 1.5) per 100,000 person-years; CVST with thrombocytopenia was only seen in SIDIAP CMBDH-HA, where the incidence rate was 0.1 (0.1 to 0.2) per 100,000 person-years. The incidence rates for SVT ranged from 1.5 (1.2 to 1.8) to 10.3 (9.8 to 10.8), and from 0.1 (0.0 to 0.1) to 2.5 (2.2 to 2.7) per 100,000 person-years for SVT with thrombocytopenia. The incidence rates for DVT ranged from 85.9 (84.6 to 87.2) to 187.2 (182.6 to 191.8), and from 1.0 (0.7 to 1.4) to 8.5 (7.4 to 9.9) per 100,000 person-years for DVT with thrombocytopenia. The incidence rates for PE ranged from 66.1 (63.0 to 69.2) to 131.2 (126.4to 136.2), and from 0.5 (0.3 to 0.6) to 20.8 (18.9 to 22.8) per 100,000 person-years for PE with thrombocytopenia. Lastly, the incidence rates for stroke ranged from 128.3 (126.0 to 130.6) to 326.6 (320.6 to 332.7), and from 0.6 (0.4 to 0.7) to 30.9 (28.6 to 33.3) per 100,000 person-years for stroke with thrombocytopenia. As with thrombosis in general, incidence rates for TTS were typically higher for older age groups, see Figure 1.

Based on the highest estimates for the overall study cohorts, one would expect approximately 1 case of CVST with thrombocytopenia, 19 of SVT with thrombocytopenia, 65 of DVT with thrombocytopenia, 159 of PE with thrombocytopenia, and 237 of stroke with thrombocytopenia among a general population of 10 million individuals per 28 days. For a cohort of the same size aged 65 or over, this would rise to 46 of SVT with thrombocytopenia, 161 of DVT with thrombocytopenia, 886 of PE with thrombocytopenia, and 616 of stroke with thrombocytopenia, see Figure 2.

The age and sex profiles of those with TTS are summarised in Table 3 and the prevalence of comorbidities and prior medication are presented in Figure 3, along with those of the study populations. The median age of the 16 individuals with CVST with thrombocytopenia in SIDIAP CMBDH-HA was 62 years old. The median age of those with DVT with thrombocytopenia ranged from 58 to 76 across the databases, from 68 to 78 for PE with thrombocytopenia, from 59 to 64 for SVT with thrombocytopenia, and from 73 to 78 for stroke with thrombocytopenia. Men generally predominated the affected cohorts, accounting for 50.0% to 71.6% of those with different TTS in the

medRxiv preprint doi: https://doi.org/10.1101/2021.05.12.21257083; this version posted September 17, 2021. The copyright holder for this preprint (which was not certified by peer review) is the author/funder, who has granted medRxiv a license to display the preprint in perpetuity.
It is made available under a CC-BY 4.0 International license .

contributing databases. The prevalence of comorbidities and prior medication use was higher for patients with TTS than in the general population. In CPRD, for example, 1.8% of the study population had an autoimmune disease, 5.1% had a history of cancer, 5.5% had diabetes, 4.3% had renal impairment. These compared to 12.6%, 25.2%, 20.5%, and 26.8% for patients with DVT with thrombocytopenia. Similarly, while 2.9% of the study population were taking antithrombotic and anticoagulant therapies in the months preceding their index date, 18.1% of patients with DVT with thrombocytopenia were. Requiring a year of prior history for study participants to be included in the analysis and defining study populations based on their first visit after 2017 had only a small effect on the results, see the Appendix.

Incidence rates for DIC, HIT, immune thrombocytopenia, and TTP are summarised in Table 4. The incidence rate for DIC ranged from 0.2 (0.1 to 0.3) to 3.8 (3.3 to 4.1) per 100,000 person-years, from 0.2 (0.1 to 0.4) to 37.9 (37.0 to 38.) for HIT, from 2.1 (1.8 to 2.5) to 46.7 (45.7 to 47.7) for immune thrombocytopenia, and from 0.4 (0.2 to 0.8) to 2.8 (2.6 to 3.1) for thrombotic thrombocytopenic purpura.

The incidence rates for all study outcomes are summarised in the supplementary materials and in a web application: https://livedataoxford.shinyapps.io/CovCoagBackgroundIncidence/, where the characteristics of outcome cohorts are also described.

Discussion

Key results

In this study we have analysed data for over 25 million people from across 6 European countries to establish the background incidence of TTS. With incidence rates of less than 35 per 100,000 person-years, TTS can be considered as a very rare event. These events can generally be expected to occur in older persons, with the average age of those over 60 for most events in most of the databases studied. Moreover, those affected typically had a higher prevalence of comorbidities, such as autoimmune diseases, cancer, and diabetes. They also had a high prevalence of use of medications indicated for the prevention of thrombosis including antithrombotic and anticoagulant therapies, as well as some potentially associated with an increased risk of TTS such as systemic glucocorticoids.

medRxiv preprint doi: https://doi.org/10.1101/2021.05.12.21257083; this version posted September 17, 2021. The copyright holder for this preprint (which was not certified by peer review) is the author/funder, who has granted medRxiv a license to display the preprint in perpetuity.
It is made available under a CC-BY 4.0 International license .

Coagulopathies potentially associated with TTS were mostly rare: immune thrombocytopenia was the most common with rates up to almost 47 per 100,000 person-years, followed by HIT (up to 38 per 100,000), DIC (up to 4 per 100,000), and TTP (up to 3 per 100,000).

Exact estimates of the background incidence of both rare events, such as CVST, and more common events, such as DVT, have though been seen to vary substantially depending on the data source used. This comes even after the use of common analytic code and with databases mapped to a common data model. More specifically, the databases which captured both inpatient and outpatient records showed the highest background rates of most of the studied events. Moreover, rates of TTS were much higher in the database which captured both inpatient and outpatient lab measurements (HIC Dundee). Indeed, while around 90% of those with VTE had a platelet count recorded in this database, only around 10% had a count in the other databases which relied on outpatient lab measurements. In both cases, around 10% of lab measurements that were observed had a value that indicated thrombocytopenia. This underlines the importance of using consistent data sources in vaccine safety research.

Findings in context

A number of previous studies have estimated the incidence of venous thromboembolism in the general population, with its incidence rate estimated to be around 100 persons per 100,000 person-years.[18] Approximately two-thirds of venous thromboembolism can be expected to present as DVT, with the other third presenting as PE with or without DVT.[19] The background of stroke has also been the subject of much research, with its incidence generally estimated to be more than 100 persons per 100,000 person-years. [20,21] The incidence of each of these events is seen to be much higher among older persons. The incidence of SVT and CVST is far less well-known. Estimates for the incidence of CVST have ranged from 0.2 to 2 per 100,000 person-years. [22–25] Meanwhile there is little research describing the incidence of SVT in the general population, although the incidence of portal vein thrombosis, the most commonly involved vein, has been estimated at around 3 per 100,000 person-years, while the incidence of Budd-Chiari syndrome was estimated at around 2 per 100,000 person-years in the same study.[26]

In a recent study data from Denmark and Norway was used to assess 28-day rates of thromboembolic events and coagulation disorders among a cohort of people who had received the ChAdOx1 vaccine and in historical comparator cohorts.[27] In the historical comparator population, which covered 2016 to 2018 for Denmark and 2018 to 2019 for Denmark, the incidence rate of CVST, PE, lower limb venous thrombosis, and SVT were estimated at 2, 57, 94, and 4 per 100,000 person-

medRxiv preprint doi: https://doi.org/10.1101/2021.05.12.21257083; this version posted September 17, 2021. The copyright holder for this preprint (which was not certified by peer review) is the author/funder, who has granted medRxiv a license to display the preprint in perpetuity.
It is made available under a CC-BY 4.0 International license .

years, respectively. Meanwhile the incidence rate for idiopathic thrombocytopenia purpura and DIC were 7 and 1 per 100,000. These estimates are broadly in line with those from our study.

The background incidence of TTS itself, however, has not been previously described in detail. Our findings demonstrate that thrombosis with concomitant thrombocytopenia is very rare, with an unadjusted overall incidence rate of less than 5 per 100,000 person-years for each of DVT, PE, CVST, SVT, and stroke with thrombocytopenia. In particular, CVST with thrombocytopenia was only seen in the database with patient-level linkage to hospital data, where its incidence rate was approximately one per million person-years. Similarly, the incidence rate for SVT with thrombocytopenia was between one and seven cases per million.

Spontaneous identified 93 cases of CVST with thrombocytopenia among individuals who had recently received the ChAdOx1 vaccine in the UK.[8] The profile of patients with TTS after vaccination also appears to differ to the typical profiles of those with TTS as seen in our data. While in this study we have seen those with TTS to typically be older than the general population, more commonly male, and with more comorbidities and greater prior medication use, initial studies describing the profiles of patients with vaccine-induced TTS have most often presented the cases of people who were aged under 60, more often female, and with relatively few comorbidities described.[28–31] While these case series of TTS after vaccination are small and their profiles may reflect the particular characteristics of those who were first to receive a vaccine against SARS-CoV-2, this dissimilarity in patient profiles of those with TTS in previous years and those for whom it has been reported following a vaccination is notable.

Where we have estimated the incidence of thrombosis in general, our findings have been very broadly in line with previous research. Estimates have, however, been seen to vary substantially across databases. This was particularly the case for estimates for DVT, for example, where a three-fold difference was seen between the databases with the highest and lowest incidence rates. Our estimates of TTS and coagulopathy also vary across the databases used. Indeed, this heterogeneity was observed although we used data mapped to a common data model and applied the same analytic code across the databases. Given that the data sources used come from different countries and can be expected to vary in completeness, accuracy, and granularity, it is not necessarily surprising that estimates vary. In addition, given the seriousness of many of the events described here, the degree to which episodes of hospital care are reflected in the data is also likely to be an important driver of the observed heterogeneity, with incidence rates higher for most of the events in the data source with linkage to hospitalisation data. The reasons for the heterogeneity seen between the databases used in this study is discussed in further detail in the Appendix.

medRxiv preprint doi: https://doi.org/10.1101/2021.05.12.21257083; this version posted September 17, 2021. The copyright holder for this preprint (which was not certified by peer review) is the author/funder, who has granted medRxiv a license to display the preprint in perpetuity.
It is made available under a CC-BY 4.0 International license .

Study limitations

This study relies on routinely-collected health care data and while this has allowed for the inclusion of a large study population, the recording of TTS has not previously been evaluated in the databases used. A degree of measurement error can thus be expected, and further research is required to validate the recording of TTS.

The degree to which the TTS events being described after vaccinations against SARS-CoV-2 are comparable to TTS events previously seen in the general population is as yet unclear. TTS after vaccination appears to occur at unusual sites, with a large proportion of spontaneous reports and case series describing cerebral or abdominal thromboses, and with high levels of antibodies to platelet factor 4 often observed despite the absence of an exposure to heparin.[28,32] In this study we have focused on specific sites of thrombosis with concomitant thrombocytopenia. We believe that this is more instructive than providing a singular background incidence rate for venous thromboembolism with thrombocytopenia, which would be driven in large part by commonly seen events (such as DVT and PE) and would not necessarily reflect the presentation of TTS after vaccination. As the pathophysiology of TTS after vaccination becomes better understood, definitions of the appropriate historical comparator can also be expected to evolve so as to best match the condition being described among those who have been recently vaccinated.

Conclusion

Based on data from over 25 million people from six European databases, TTS has been seen to be very rare. While rates varied across databases, the highest incidence rates for DVT, PE, and stroke with thrombocytopenia were 8.5, 20.8, and 30.9 per 100,000 person-years, respectively. Meanwhile the highest incidence rates for CVST and SVT with thrombocytopenia were 0.1 and 2.5 per 100,000 person-years. TTS was typically seen among individuals older, more often male, and in worse health than the general population. While these findings help to provide context for the rates of adverse events being reported by spontaneous reports following vaccinations against SARS-CoV-2, a full assessment of the safety signal for TTS would benefit from within-database comparisons which account for individual-level characteristics such as age and sex.

medRxiv preprint doi: https://doi.org/10.1101/2021.05.12.21257083; this version posted September 17, 2021. The copyright holder for this preprint (which was not certified by peer review) is the author/funder, who has granted medRxiv a license to display the preprint in perpetuity.
It is made available under a CC-BY 4.0 International license .

References

1. Krammer, F. SARS-CoV-2 vaccines in development. *Nature* **586**, 516–527 (2020).

2. Baden, L. R. *et al.* Efficacy and Safety of the mRNA-1273 SARS-CoV-2 Vaccine. *N. Engl. J. Med.* **384**, 403–416 (2020).

3. Polack, F. P. *et al.* Safety and Efficacy of the BNT162b2 mRNA Covid-19 Vaccine. *N. Engl. J. Med.* **383**, 2603–2615 (2020).

4. Voysey, M. *et al.* Safety and efficacy of the ChAdOx1 nCoV-19 vaccine (AZD1222) against SARS-CoV-2: an interim analysis of four randomised controlled trials in Brazil, South Africa, and the UK. *Lancet* **397**, 99–111 (2021).

5. Nature. 'Unprecedented achievement': who received the first billion COVID vaccinations? (2021). doi:https://doi.org/10.1038/d41586-021-01136-2

6. Dagan, N. *et al.* BNT162b2 mRNA Covid-19 Vaccine in a Nationwide Mass Vaccination Setting. *N. Engl. J. Med.* **384**, 1412–1423 (2021).

7. Vasileiou, E. *et al.* Interim findings from first-dose mass COVID-19 vaccination roll-out and COVID-19 hospital admissions in Scotland: a national prospective cohort study. *Lancet* **397**, 1646–1657 (2021).

8. MHRA. Coronavirus vaccine - weekly summary of Yellow Card reporting: Updated 6 May 2021. (2021). Available at: https://www.gov.uk/government/publications/coronavirus-covid-19-vaccine-adverse-reactions/coronavirus-vaccine-summary-of-yellow-card-reporting#analysis-of-data. (Accessed: 7th May 2021)

9. CDC. J&J/Janssen Update. (2021). Available at: https://www.cdc.gov/coronavirus/2019-ncov/vaccines/safety/JJUpdate.html. (Accessed: 7th May 2021)

10. Black, S. *et al.* Importance of background rates of disease in assessment of vaccine safety during mass immunisation with pandemic H1N1 influenza vaccines. *Lancet* **374**, 2115–2122 (2009).

11. Voss, E. A. *et al.* Feasibility and utility of applications of the common data model to multiple, disparate observational health databases. *J Am Med Inf. Assoc* **22**, 553–564 (2015).

medRxiv preprint doi: https://doi.org/10.1101/2021.05.12.21257083; this version posted September 17, 2021. The copyright holder for this preprint (which was not certified by peer review) is the author/funder, who has granted medRxiv a license to display the preprint in perpetuity.
It is made available under a CC-BY 4.0 International license .

12. Hripcsak, G. *et al.* Observational Health Data Sciences and Informatics (OHDSI): Opportunities for Observational Researchers. *Stud. Health Technol. Inform.* **216**, 574–8 (2015).

13. Overhage, J. M., Ryan, P. B., Reich, C. G., Hartzema, A. G. & Stang, P. E. Validation of a common data model for active safety surveillance research. *J. Am. Med. Informatics Assoc.* **19**, 54–60 (2011).

14. Herrett, E. *et al.* Data Resource Profile: Clinical Practice Research Datalink (CPRD). *Int. J. Epidemiol.* **44**, 827–836 (2015).

15. Jouaville, S. L., Miotti, H., Coffin, G., Sarfati, B. & Meihoc, A. Validity And Limitations Of The Longitudinal Patient Database France For Use In Pharmacoepidemiological And Pharmacoeconomics Studies. *Value Heal.* **18**, A18 (2015).

16. García-Gil, M. D. M. *et al.* Construction and validation of a scoring system for the selection of high-quality data in a Spanish population primary care database (SIDIAP). *Inform. Prim. Care* **19**, 135–45 (2011).

17. Schuemie, M., Ryan, P. & Weaver, J. CohortDiagnostics: Diagnostics for OHDSI studies. (2020).

18. H., W. R. The Epidemiology of Venous Thromboembolism. *Circulation* **107**, I-4-I–8 (2003).

19. Cushman, M. Epidemiology and Risk Factors for Venous Thrombosis. *Semin. Hematol.* **44**, 62–69 (2007).

20. Feigin, V. L., Lawes, C. M. M., Bennett, D. A., Barker-Collo, S. L. & Parag, V. Worldwide stroke incidence and early case fatality reported in 56 population-based studies: a systematic review. *Lancet Neurol.* **8**, 355–369 (2009).

21. Krishnamurthi, R. V *et al.* Global and regional burden of first-ever ischaemic and haemorrhagic stroke during 1990–2010: findings from the Global Burden of Disease Study 2010. *Lancet Glob. Heal.* **1**, e259–e281 (2013).

22. Otite, F. O. *et al.* Trends in incidence and epidemiologic characteristics of cerebral venous thrombosis in the United States. *Neurology* **95**, e2200 LP-e2213 (2020).

23. Stam, J. Thrombosis of the Cerebral Veins and Sinuses. *N. Engl. J. Med.* **352**, 1791–1798 (2005).

medRxiv preprint doi: https://doi.org/10.1101/2021.05.12.21257083; this version posted September 17, 2021. The copyright holder for this preprint **(which was not certified by peer review)** is the author/funder, who has granted medRxiv a license to display the preprint in perpetuity.
It is made available under a CC-BY 4.0 International license .

24. M., C. J., M., Z. S., Majid, A. & Jan, S. The Incidence of Cerebral Venous Thrombosis. *Stroke* **43**, 3375–3377 (2012).

25. Sharon, D., Ben, W., James, L. & Timothy, K. Cerebral Venous Sinus Thrombosis Incidence Is Higher Than Previously Thought. *Stroke* **47**, 2180–2182 (2016).

26. Ageno, W. *et al.* Incidence rates and case fatality rates of portal vein thrombosis and Budd-Chiari Syndrome. *Thromb Haemost* **117**, 794–800 (2017).

27. Pottegård, A. *et al.* Arterial events, venous thromboembolism, thrombocytopenia, and bleeding after vaccination with Oxford-AstraZeneca ChAdOx1-S in Denmark and Norway: population based cohort study. *BMJ* **373**, n1114 (2021).

28. Cines, D. B. & Bussel, J. B. SARS-CoV-2 Vaccine–Induced Immune Thrombotic Thrombocytopenia. *N. Engl. J. Med.* (2021). doi:10.1056/NEJMe2106315

29. Schultz, N. H. *et al.* Thrombosis and Thrombocytopenia after ChAdOx1 nCoV-19 Vaccination. *N. Engl. J. Med.* (2021). doi:10.1056/NEJMoa2104882

30. Scully, M. *et al.* Pathologic Antibodies to Platelet Factor 4 after ChAdOx1 nCoV-19 Vaccination. *N. Engl. J. Med.* (2021). doi:10.1056/NEJMoa2105385

31. See, I. *et al.* US Case Reports of Cerebral Venous Sinus Thrombosis With Thrombocytopenia After Ad26.COV2.S Vaccination, March 2 to April 21, 2021. *JAMA* (2021). doi:10.1001/jama.2021.7517

32. Karron, R. A., Key, N. S. & Sharfstein, J. M. Assessing a Rare and Serious Adverse Event Following Administration of the Ad26.COV2.S Vaccine. *JAMA* (2021). doi:10.1001/jama.2021.7637

medRxiv preprint doi: https://doi.org/10.1101/2021.05.12.21257083; this version posted September 17, 2021. The copyright holder for this preprint (which was not certified by peer review) is the author/funder, who has granted medRxiv a license to display the preprint in perpetuity.
It is made available under a CC-BY 4.0 International license .

Ethical approvals

The protocol for this research was approved by the Independent Scientific Advisory Committee (ISAC) for MHRA Database Research (protocol number 20_000211), the IDIAPJGol Clinical Research Ethics Committee (project code: 21/007-PCV), and the IPCI governance board (application number 3/2021). Some databases used (IQVIA LPD Italy, IQVIA LPD France, IQVIA DA Germany) in these analyses are commercially available, syndicated data assets that are licensed by contributing authors for observational research. These assets are de-identified commercially available data products that could be purchased and licensed by any researcher. As these data are deemed commercial assets, there is no Institutional Review Board applicable to the usage and dissemination of these result sets or required registration of the protocol with additional ethics oversight. Compliance with Data Use Agreement terms, which stipulate how these data can be used and for what purpose, is sufficient for the licensing commercial entities. Further inquiry related to the governance oversight of these assets can be made with the respective commercial entity, IQVIA (iqvia.com). For HIC Dundee, institutional review board approval for the use of de-identified data for this project was granted by the Tayside Health Informatics Centre.

Acknowledgements

This study was funded by the European Medicines Agency in the form of a competitive tender (Lot ROC No EMA/2017/09/PE). We acknowledge Prof Johan Van der Lei for the overall management of this research grant.

Declarations of interest

DPA's research group has received research grants from the European Medicines Agency, from the Innovative Medicines Initiative, from Amgen, Chiesi, and from UCB Biopharma; and consultancy or speaker fees from Astellas, Amgen and UCB Biopharma. At the time of analysis, KK, HMS, CR and SS were employees of IQVIA. KK reported receiving funding from the National Institutes of Health National COVID Cohort Collaborative (N3C). IQVIA received funding from the University of Oxford on behalf of the Bill & Melinda Gates Foundation for the conversion of LPD Italy and utilization of DA Germany data for COVID-19 related research.

medRxiv preprint doi: https://doi.org/10.1101/2021.05.12.21257083; this version posted September 17, 2021. The copyright holder for this preprint (which was not certified by peer review) is the author/funder, who has granted medRxiv a license to display the preprint in perpetuity.
It is made available under a CC-BY 4.0 International license .

Author contributions

All authors were involved in the study conception and design, interpretation of the results, and the preparation of the manuscript. EB led the data analysis and wrote the initial draft of the manuscript with DPA. EB, TDS, CR, MA, and SFB had access to the SIDIAP data, EB, XL, AD, and DPA had access to the CPRD data, DM and SH had access to the HIC Dundee data, PR had access to the IPCI data, and KK, HMS, CR, SS, had access to LPD France, LPD Italy, and DA Germany.

Tables and Figures

Table 1: Characteristics of study populations

	CPRD	**HIC Dundee**	**IPCI The Netherlands**	**IQVIA DA Germany**	**IQVIA LPD France**	**IQVIA LPD Italy**	**SIDIAP CMBDH-HA**
N	3,913,071	948,561	1,299,288	8,459,098	3,951,633	1,066,230	5,794,777
Age (Median [IQR])	41 [22 to 59]	41 [23 to 59]	44 [23 to 60]	52 [32 to 67]	48 [28 to 65]	52 [37 to 68]	42 [25 to 59]
Sex: Male (N [%])	1,937,858 (49.5%)	469,725 (49.5%)	636,386 (49.0%)	3,589,506 (42.4%)	1,669,415 (42.2%)	426,758 (40.0%)	2,859,044 (49.3%)
Years of prior observation time (Median [IQR])	11.9 [4.7 to 15.1]	8.0 [6.6 to 12.0]	3.2 [1.8 to 5.7]	4.8 [1.9 to 8.9]	4.6 [2.0 to 6.2]	6.3 [5.0 to 6.5]	11.0 [11.0 to 11.0]
Comorbidities							
Autoimmune disease (N [%])	70,604 (1.8%)	8,040 (0.8%)	24,645 (1.9%)	238,985 (2.8%)	32,245 (0.8%)	45,567 (4.3%)	84,817 (1.5%)
Antiphospholipid syndrome (N [%])	1,166 (0.0%)	<5	<5	<5	<5	<5	1,011 (0.0%)

Thrombophilia (N [%])	3,039 (0.1%)	198 (0.0%)	0 (0.0%)	6,474 (0.1%)	313 (0.0%)	<5	2,796 (0.0%)
Asthma (N [%])	484,991 (12.4%)	37,160 (3.9%)	138,777 (10.7%)	412,789 (4.9%)	222,161 (5.6%)	79,528 (7.5%)	353,485 (6.1%)
COPD (N [%])	80,393 (2.1%)	14,225 (1.5%)	40,116 (3.1%)	358,047 (4.2%)	41,040 (1.0%)	27,119 (2.5%)	166,817 (2.9%)
Atrial fibrillation (N [%])	76,091 (1.9%)	825 (0.1%)	31,801 (2.4%)	92,767 (1.1%)	13,412 (0.3%)	34,325 (3.2%)	137,843 (2.4%)
Diabetes mellitus (N [%])	213,996 (5.5%)	25,891 (2.7%)	93,035 (7.2%)	597,233 (7.1%)	174,564 (4.4%)	95,611 (9.0%)	468,808 (8.1%)
Obesity (N [%])	107,522 (2.7%)	9,843 (1.0%)	40,395 (3.1%)	530,958 (6.3%)	15,634 (0.4%)	46,101 (4.3%)	927,483 (16.0%)
Heart disease (N [%])	278,323 (7.1%)	56,966 (6.0%)	129,562 (10.0%)	936,730 (11.1%)	194,630 (4.9%)	165,172 (15.5%)	592,122 (10.2%)
Hypertensive disorder (N [%])	558,671 (14.3%)	71,703 (7.6%)	222,433 (17.1%)	1,425,782 (16.9%)	498,244 (12.6%)	322,776 (30.3%)	1,145,518 (19.8%)
Renal impairment (N [%])	168,610 (4.3%)	17,311 (1.8%)	27,555 (2.1%)	169,166 (2.0%)	13,064 (0.3%)	31,853 (3.0%)	230,896 (4.0%)
Malignant neoplastic disease (N [%])	198,275 (5.1%)	51,307 (5.4%)	106,223 (8.2%)	534,352 (6.3%)	66,962 (1.7%)	86,645 (8.1%)	342,511 (5.9%)

Dementia (N [%])	33,537 (0.9%)	5,515 (0.6%)	7,873 (0.6%)	95,957 (1.1%)	9,217 (0.2%)	10,458 (1.0%)	72,696 (1.3%)
Medication use (183 days prior to four days prior)							
Non-steroidal anti-inflammatory drugs (N [%])	900,092 (23.0%)	247,182 (26.1%)	211,464 (16.3%)	928,497 (11.0%)	1,056,021 (26.7%)	293,188 (27.5%)	1,617,103 (27.9%)
Cox-2 inhibitors (N [%])	7,126 (0.2%)	2,469 (0.3%)	8,165 (0.6%)	33,006 (0.4%)	13,769 (0.3%)	21,899 (2.1%)	27,048 (0.5%)
Systemic corticosteroids (N [%])	404,443 (10.3%)	95,032 (10.0%)	139,482 (10.7%)	269,020 (3.2%)	315,054 (8.0%)	84,587 (7.9%)	337,121 (5.8%)
Antithrombotic and anticoagulant therapies (N [%])	114,246 (2.9%)	72,557 (7.6%)	44,985 (3.5%)	213,378 (2.5%)	175,535 (4.4%)	110,079 (10.3%)	112,901 (1.9%)
Lipid modifying agents (N [%])	143,424 (3.7%)	103,695 (10.9%)	54,039 (4.2%)	191,407 (2.3%)	197,031 (5.0%)	83,234 (7.8%)	81,743 (1.4%)
Antineoplastic and immunomodulating agents (N [%])	124,080 (3.2%)	44,853 (4.7%)	54,941 (4.2%)	210,390 (2.5%)	94,702 (2.4%)	36,792 (3.5%)	64,163 (1.1%)
Hormonal contraceptives for systemic use (N [%])	173,708 (4.4%)	51,084 (5.4%)	47,983 (3.7%)	169,549 (2.0%)	98,852 (2.5%)	18,740 (1.8%)	46,834 (0.8%)
Tamoxifen (N [%])	2,141 (0.1%)	1,666 (0.2%)	865 (0.1%)	3,761 (0.0%)	826 (0.0%)	684 (0.1%)	1,230 (0.0%)

Sex hormones and modulators of the genital system (N [%])	213,023 (5.4%)	63,019 (6.6%)	55,810 (4.3%)	228,846 (2.7%)	141,501 (3.6%)	29,750 (2.8%)	58,987 (1.0%)

CPRD: Clinical Practice Research Datalink, IQVIA DA GERMANY: IQVIA Disease Analyser Germany, IQVIA LPD France: IQVIA Longitudinal Patient Data France, IPCI: Integrated Primary Care Information, IQVIA LPD Italy: IQVIA Longitudinal Patient Data Italy, SIDIAP CMBDH-HA: Information System for Research in Primary Care with hospital linkage.

COPD: chronic obstructive pulmonary disease

Table 2. Incidence rates per 100,000 person-years for thrombosis and thrombosis with thrombocytopenia

	N	**PYs**	**Number of events**	**Incidence rate (95% CI) per 100,000 PYs**
Cerebral venous sinus thrombosis (CVST)				
CPRD	3,913,025	9,676,085	118	1.2 (1.0 to 1.5)
IQVIA DA Germany	8,459,044	19,369,671	95	0.5 (0.4 to 0.6)
IQVIA LPD France	3,951,606	8,210,128	26	0.3 (0.2 to 0.5)
SIDIAP CMBDH-HA	5,794,764	16,751,651	121	0.7 (0.6 to 0.9)
Cerebral venous sinus thrombosis (CVST) with thrombocytopenia				
SIDIAP CMBDH-HA	5,794,777	16,751,791	16	0.1 (0.1 to 0.2)
Deep vein thrombosis (DVT)				
CPRD	3,909,649	9,656,721	9,071	93.9 (92.0 to 95.9)
HIC Dundee	948,184	2,153,442	1,186	55.1 (52.0 to 58.3)

IQVIA DA Germany	8,451,032	19,329,175	16,600	85.9 (84.6 to 87.2)
IPCI	1,296,310	3,402,027	6,367	187.2 (182.6 to 191.8)
IQVIA LPD Italy	1,063,587	2,639,975	3,891	147.4 (142.8 to 152.1)
SIDIAP CMBDH-HA	5,790,802	16,722,842	14408	86.2 (84.8 to 87.6)
Deep vein thrombosis (DVT) with thrombocytopenia				
CPRD	3,913,031	9,676,132	127	1.3 (1.1 to 1.6)
HIC Dundee	948,498	2,153,995	184	8.5 (7.4 to 9.9)
IQVIA DA Germany	8,458,995	19,369,390	225	1.2 (1.0 to 1.3)
IPCI	1,299,274	3,418,833	34	1.0 (0.7 to 1.4)
IQVIA LPD Italy	1,066,209	2,651,714	39	1.5 (1.0 to 2.0)
SIDIAP CMBDH-HA	5,794,559	16,750,224	1037	6.2 (5.8 to 6.6)
Pulmonary embolism (PE)				

CPRD	3,910,531	9,662,585	7149	74.0 (72.3 to 75.7)
HIC Dundee	947,984	2,151,351	2823	131.2 (126.4 to 136.2)
IQVIA DA Germany	8,449,246	19,325,600	17204	89.0 (87.7 to 90.4)
IQVIA LPD France	3,947,450	8,195,137	4700	57.4 (55.7 to 59.0)
IPCI	1,297,807	3,410,984	3141	92.1 (88.9 to 95.4)
IQVIA LPD Italy	1,064,532	2,645,601	1748	66.1 (63.0 to 69.2)
SIDIAP CMBDH-HA	5,792,195	16,734,624	9590	57.3 (56.2 to 58.5)
Pulmonary embolism (PE) with thrombocytopenia				
CPRD	3,913,042	9,676,256	84	0.9 (0.7 to 1.1)
HIC Dundee	948,459	2,153,685	447	20.8 (18.9 to 22.8)
DA Germany	8,458,971	19,369,265	286	1.5 (1.3 to 1.7)
IQVIA LPD France	3,951,605	8,210,109	39	0.5 (0.3 to 0.6)

IPCI	1,299,282	3,418,860	21	0.6 (0.4 to 0.9)
IQVIA LPD Italy	1,066,222	2,651,761	17	0.6 (0.4 to 1.0)
SIDIAP CMBDH-HA	5,794,594	16,750,449	985	5.9 (5.5 to 6.3)
Splanchnic vein thrombosis (SVT)				
CPRD	3,913,005	9,675,960	233	2.4 (2.1 to 2.7)
HIC Dundee	948,541	2,154,020	112	5.2 (4.3 to 6.3)
IQVIA DA Germany	8,458,941	19,369,177	398	2.1 (1.9 to 2.3)
IQVIA LPD France	3,951,594	8,210,016	122	1.5 (1.2 to 1.8)
IQVIA LPD Italy	1,066,207	2,651,684	58	2.2 (1.7 to 2.8)
SIDIAP CMBDH-HA	5,794,483	16,749,703	1718	10.3 (9.8 to 10.8)
Splanchnic vein thrombosis (SVT) with thrombocytopenia				
CPRD	3,913,070	9,676,375	5	0.1 (0.0 to 0.1)

HIC Dundee	948,553	2,154,098	40	1.9 (1.3 to 2.5)
IQVIA DA Germany	8,459,086	19,369,887	16	0.1 (0.0 to 0.1)
SIDIAP CMBDH-HA	5,794,721	16,751,354	412	2.5 (2.2 to 2.7)
Stroke				
CPRD	3,908,602	9,650,786	12381	128.3 (126.0 to 130.6)
HIC Dundee	947,200	2,146,633	6,326	294.7 (287.5 to 302.0)
IQVIA DA Germany	8,434,716	19,257,348	46784	242.9 (240.7 to 245.2)
IQVIA LPD France	3,944,166	8,178,421	12899	157.7 (155.0 to 160.5)
IPCI	1,293,518	3,390,322	11073	326.6 (320.6 to 332.7)
SIDIAP CMBDH-HA	5,782,776	16,671,995	42126	252.7 (250.3 to 255.1)
Stroke with thrombocytopenia				
CPRD	3,913,039	9,676,223	79	0.8 (0.6 to 1.0)

HIC Dundee	948,431	2,153,504	665	30.9 (28.6 to 33.3)
IQVIA DA Germany	8,458,923	19,369,062	369	1.9 (1.7 to 2.1)
IQVIA LPD France	3,951,616	8,210,137	46	0.6 (0.4 to 0.7)
IPCI	1,299,278	3,418,845	32	0.9 (0.6 to 1.3)
SIDIAP CMBDH-HA	5,794,204	16,747,541	3058	18.3 (17.6 to 18.9)

CPRD: Clinical Practice Research Datalink, IQVIA DA GERMANY: IQVIA Disease Analyser Germany, IQVIA LPD France: IQVIA Longitudinal Patient Data France, IPCI: Integrated Primary Care Information, IQVIA LPD Italy: IQVIA Longitudinal Patient Data Italy, SIDIAP CMBDH-HA: Information System for Research in Primary Care with hospital linkage.

Table 3. Characteristics of patients with thrombosis with thrombocytopenia

	N	**Age (Median [IQR])**	**Sex: Male (N [%])**
CPRD			
Study population	3,913,071	41 [22 to 59]	1,937,858 (49.5%)
Deep vein thrombosis with thrombocytopenia	127	70 [56 to 80]	70 (55.1%)
Pulmonary embolism with thrombocytopenia	84	71 [62 to 79]	50 (59.5%)
Splanchnic vein thrombosis with thrombocytopenia	5	59 [52 to 59]	<5
Stroke with thrombocytopenia	79	74 [66 to 81]	53 (67.1%)
HIC Dundee			
Study population	948,561	41 [23 to 59]	469,725 (49.5%)

Deep vein thrombosis with thrombocytopenia	184	58 [37 to 75]	99 (53.8%)
Pulmonary embolism with thrombocytopenia	447	68 [53 to 78]	247 (55.3%)
Splanchnic vein thrombosis with thrombocytopenia	40	65 [52 to 72]	27 (67.5%)
Stroke with thrombocytopenia	665	77 [67 to 84]	398 (59.8%)
IQVIA DA Germany			
Study population	8,459,098	52 [32 to 67]	3,589,506 (42.4%)
Deep vein thrombosis with thrombocytopenia	225	71 [60 to 80]	143 (63.6%)
Pulmonary embolism with thrombocytopenia	286	72 [62 to 80]	183 (64.0%)
Splanchnic vein thrombosis with thrombocytopenia	16	64 [60 to 73]	11 (68.8%)

Stroke with thrombocytopenia	369	76 [66 to 82]	244 (66.1%)
IQVIA LPD France			
Study population	3,951,633	48 [28 to 65]	1,669,415 (42.2%)
Pulmonary embolism with thrombocytopenia	39	72 [57 to 83]	23 (59.0%)
Stroke with thrombocytopenia	46	73 [66 to 82]	30 (65.2%)
IPCI			
Study population	1,299,288	44 [23 to 60]	636,386 (49.0%)
Deep vein thrombosis with thrombocytopenia	34	70 [54 to 81]	20 (58.8%)
Pulmonary embolism with thrombocytopenia	21	70 [54 to 73]	12 (57.1%)

Stroke with thrombocytopenia	32	74 [68 to 85]	22 (68.8%)
IQVIA LPD Italy			
Study population	1,066,230	52 [37 to 68]	426,758 (40.0%)
Deep vein thrombosis with thrombocytopenia	39	76 [62 to 82]	20 (51.3%)
Pulmonary embolism with thrombocytopenia	17	78 [69 to 81]	9 (52.9%)
SIDIAP CMBDH-HA			
Study population	5,794,777	42 [25 to 59]	2,859,044 (49.3%)
Cerebral venous sinus thrombosis with thrombocytopenia	16	62 [49 to 67]	8 (50.0%)
Deep vein thrombosis with thrombocytopenia	1,037	69 [58 to 79]	617 (59.5%)

Pulmonary embolism with thrombocytopenia	985	70 [59 to 79]	584 (59.3%)
Splanchnic vein thrombosis with thrombocytopenia	412	62 [54 to 72]	295 (71.6%)
Stroke with thrombocytopenia	3,058	76 [67 to 83]	2,005 (65.6%)

CPRD: Clinical Practice Research Datalink, IQVIA DA GERMANY: IQVIA Disease Analyser Germany, IQVIA LPD France: IQVIA Longitudinal Patient Data France, IPCI: Integrated Primary Care Information, IQVIA LPD Italy: IQVIA Longitudinal Patient Data Italy, SIDIAP CMBDH-HA: Information System for Research in Primary Care with hospital linkage

Table 4. Incidence rates per 100,000 person-years for coagulopathy

	N	PYs	Number of events	Incidence rate per 100, 000 PYs
Disseminated intravascular coagulation				
CPRD	3,913,067	9,676,361	15	0.2 (0.1 to 0.3)
HIC Dundee	948,555	2,154,105	32	1.5 (1.0 to 2.1)
IQVIA DA Germany	8,459,041	19,369,706	79	0.4 (0.3 to 0.5)
IQVIA France LPD	3,951,611	8,210,148	34	0.4 (0.3 to 0.6)
IQVIA Italy LPD	1,066,206	2,651,697	37	1.4 (1.0 to 1.9)
SIDIAP CMBDH-HA	5,794,596	16,750,823	641	3.8 (3.5 to 4.1)
Heparin-induced thrombocytopenia				
CPRD	3,912,943	9,675,822	302	3.1 (2.8 to 3.5)

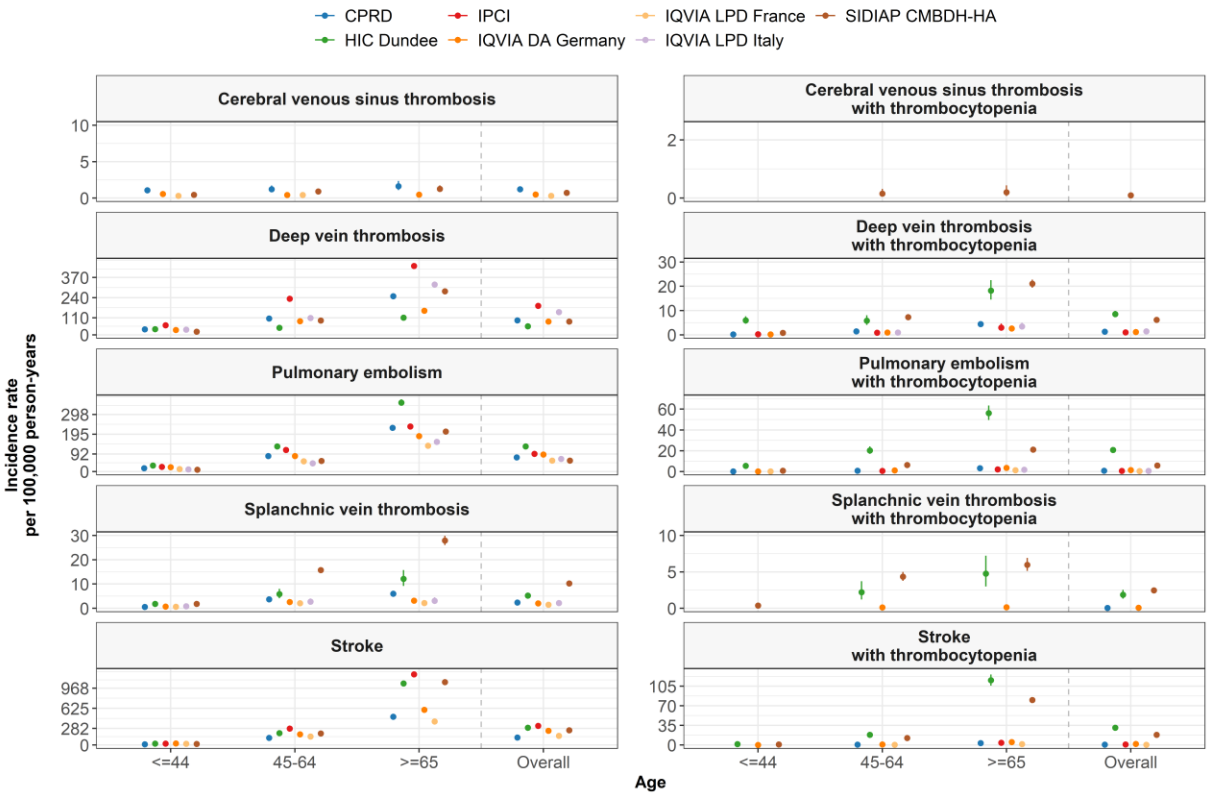
HIC Dundee	948,500	2,153,919	165	7.7 (6.5 to 8.9)
IQVIA DA Germany	8,458,456	19,366,573	1513	7.8 (7.4 to 8.2)
IQVIA France LPD	3,951,623	8,210,192	20	0.2 (0.1 to 0.4)
IQVIA Italy LPD	1,066,144	2,651,425	171	6.4 (5.5 to 7.5)
SIDIAP CMBDH-HA	5,792,945	16,739,615	6351	37.9 (37.0 to 38.9)
Immune thrombocytopenia				
CPRD	3,912,708	9,674,616	759	7.8 (7.3 to 8.4)
HIC Dundee	948,447	2,153,668	333	15.5 (13.8 to 17.2)
IQVIA DA Germany	8,457,949	19,364,321	2264	11.7 (11.2 to 12.2)
IQVIA LPD France	3,951,527	8,209,807	175	2.1 (1.8 to 2.5)
IPCI	1,299,133	3,418,075	267	7.8 (6.9 to 8.8)

SIDIAP CMBDH-HA	5,792,354	16,736,041	7816	46.7 (45.7 to 47.7)
Thrombotic thrombocytopenic purpura				
CPRD	3,913,059	9,676,289	52	0.5 (0.4 to 0.7)
HIC Dundee	948,560	2,154,123	9	0.4 (0.2 to 0.8)
IQVIA DA Germany	8,458,805	19,368,465	552	2.8 (2.6 to 3.1)
IQVIA LPD France	3,951,599	8,210,086	52	0.6 (0.5 to 0.8)
IQVIA LPD Italy	1,066,176	2,651,585	45	1.7 (1.2 to 2.3)
SIDIAP CMBDH-HA	5,794,674	16,751,190	272	1.6 (1.4 to 1.8)

CPRD: Clinical Practice Research Datalink, IQVIA DA GERMANY: IQVIA Disease Analyser Germany, IQVIA LPD France: IQVIA Longitudinal Patient Data France, IPCI: Integrated Primary Care Information, IQVIA LPD Italy: IQVIA Longitudinal Patient Data Italy, SIDIAP CMBDH-HA: Information System for Research in Primary Care with hospital linkage

medRxiv preprint doi: https://doi.org/10.1101/2021.05.12.21257083; this version posted September 17, 2021. The copyright holder for this preprint **(which was not certified by peer review)** is the author/funder, who has granted medRxiv a license to display the preprint in perpetuity.
It is made available under a CC-BY 4.0 International license .

Figure 1. Incidence rates (with 95% confidence intervals) per 100,000 of TTS among the general population, stratified by age and sex



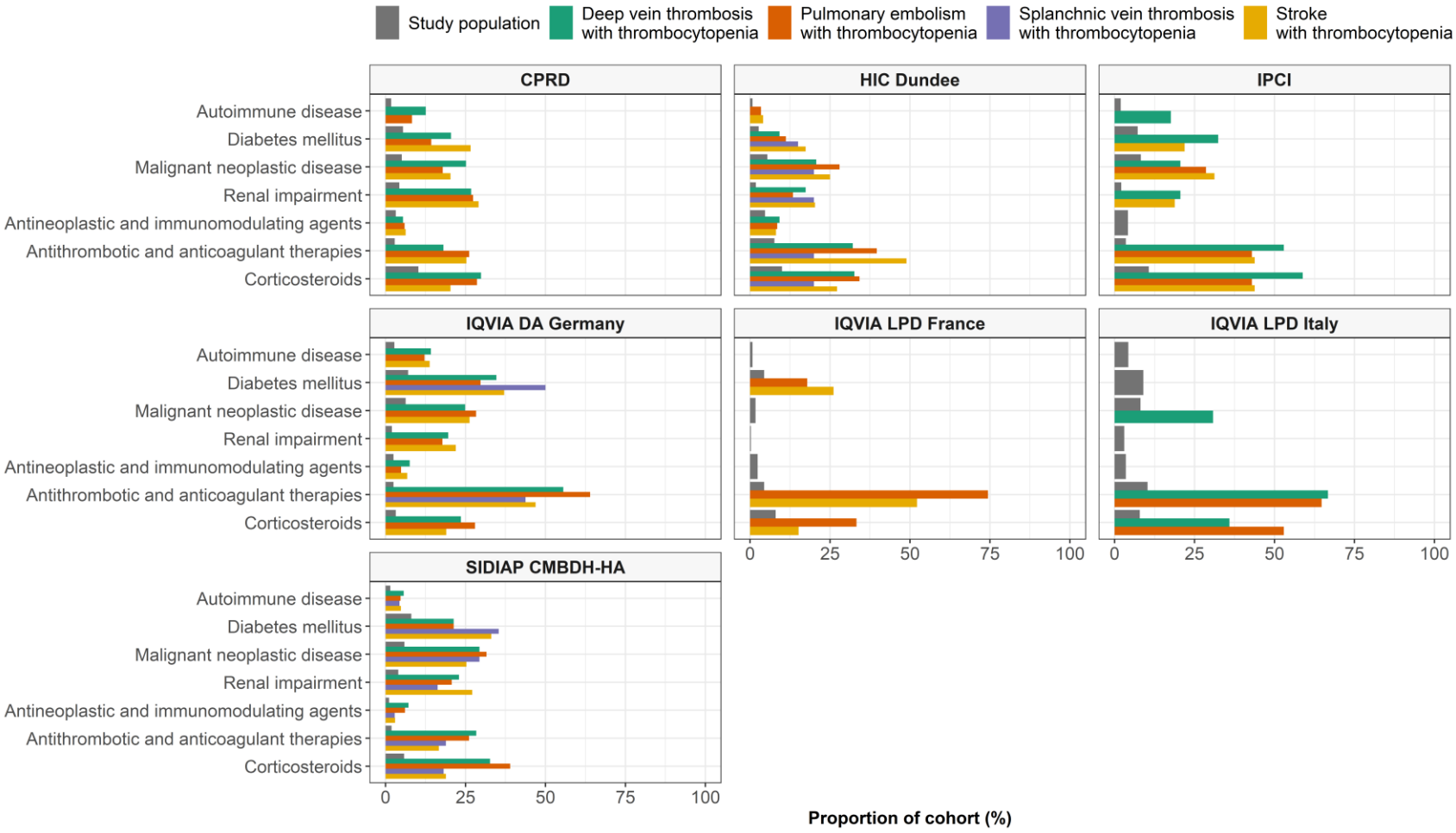
medRxiv preprint doi: https://doi.org/10.1101/2021.05.12.21257083; this version posted September 17, 2021. The copyright holder for this preprint **(which was not certified by peer review)** is the author/funder, who has granted medRxiv a license to display the preprint in perpetuity.
It is made available under a CC-BY 4.0 International license .

Figure 2. Expected cases (with 95% confidence intervals) of TTS per 28 days in a population of 10,000,000 people in a given age strata or overall. Blank cells are where there were fewer than five people with the event and incidence rates were not estimated.

	Age <=44	Age 45-64	Age >=65	Overall	
Cerebral venous sinus thrombosis with thrombocytopenia		1 (0 to 2)	2 (1 to 3)	1 (0 to 1)	SIDIAP CMBDH-HA
Deep vein thrombosis with thrombocytopenia	2 (1 to 3)	11 (8 to 15)	34 (27 to 42)	10 (8 to 12)	CPRD
	46 (35 to 58)	45 (31 to 62)	140 (111 to 173)	65 (56 to 76)	HIC Dundee
	2 (1 to 5)	7 (3 to 13)	23 (14 to 36)	8 (5 to 11)	IPCI
	2 (1 to 3)	7 (6 to 9)	20 (17 to 24)	9 (8 to 10)	IQVIA DA Germany
		8 (3 to 14)	27 (18 to 38)	11 (8 to 15)	IQVIA LPD Italy
	6 (5 to 8)	56 (50 to 62)	161 (149 to 174)	47 (45 to 50)	SIDIAP CMBDH-HA
Pulmonary embolism with thrombocytopenia	1 (0 to 2)	6 (4 to 9)	25 (19 to 32)	7 (5 to 8)	CPRD
	43 (33 to 55)	156 (130 to 185)	430 (380 to 486)	159 (145 to 175)	HIC Dundee
		5 (2 to 10)	16 (9 to 27)	5 (3 to 7)	IPCI
	2 (1 to 3)	9 (7 to 11)	27 (24 to 31)	11 (10 to 13)	IQVIA DA Germany
	2 (1 to 3)		10 (7 to 15)	4 (3 to 5)	IQVIA LPD France
			14 (8 to 23)	5 (3 to 8)	IQVIA LPD Italy
	6 (4 to 7)	48 (43 to 54)	161 (149 to 174)	45 (42 to 48)	SIDIAP CMBDH-HA
Splanchnic vein thrombosis with thrombocytopenia				0 (0 to 1)	CPRD
		17 (9 to 28)	37 (23 to 55)	14 (10 to 19)	HIC Dundee
		1 (0 to 2)	1 (0 to 2)	1 (0 to 1)	IQVIA DA Germany
	3 (2 to 4)	33 (29 to 38)	46 (39 to 53)	19 (17 to 21)	SIDIAP CMBDH-HA
Stroke with thrombocytopenia		5 (3 to 8)	26 (20 to 33)	6 (5 to 8)	CPRD
	12 (7 to 19)	140 (116 to 168)	886 (813 to 965)	237 (219 to 255)	HIC Dundee
			30 (20 to 44)	7 (5 to 10)	IPCI
	2 (1 to 3)	7 (6 to 10)	40 (35 to 45)	15 (13 to 16)	IQVIA DA Germany
		3 (1 to 6)	13 (9 to 18)	4 (3 to 6)	IQVIA LPD France
	8 (7 to 10)	95 (87 to 103)	616 (592 to 642)	140 (135 to 145)	SIDIAP CMBDH-HA

250 500 750

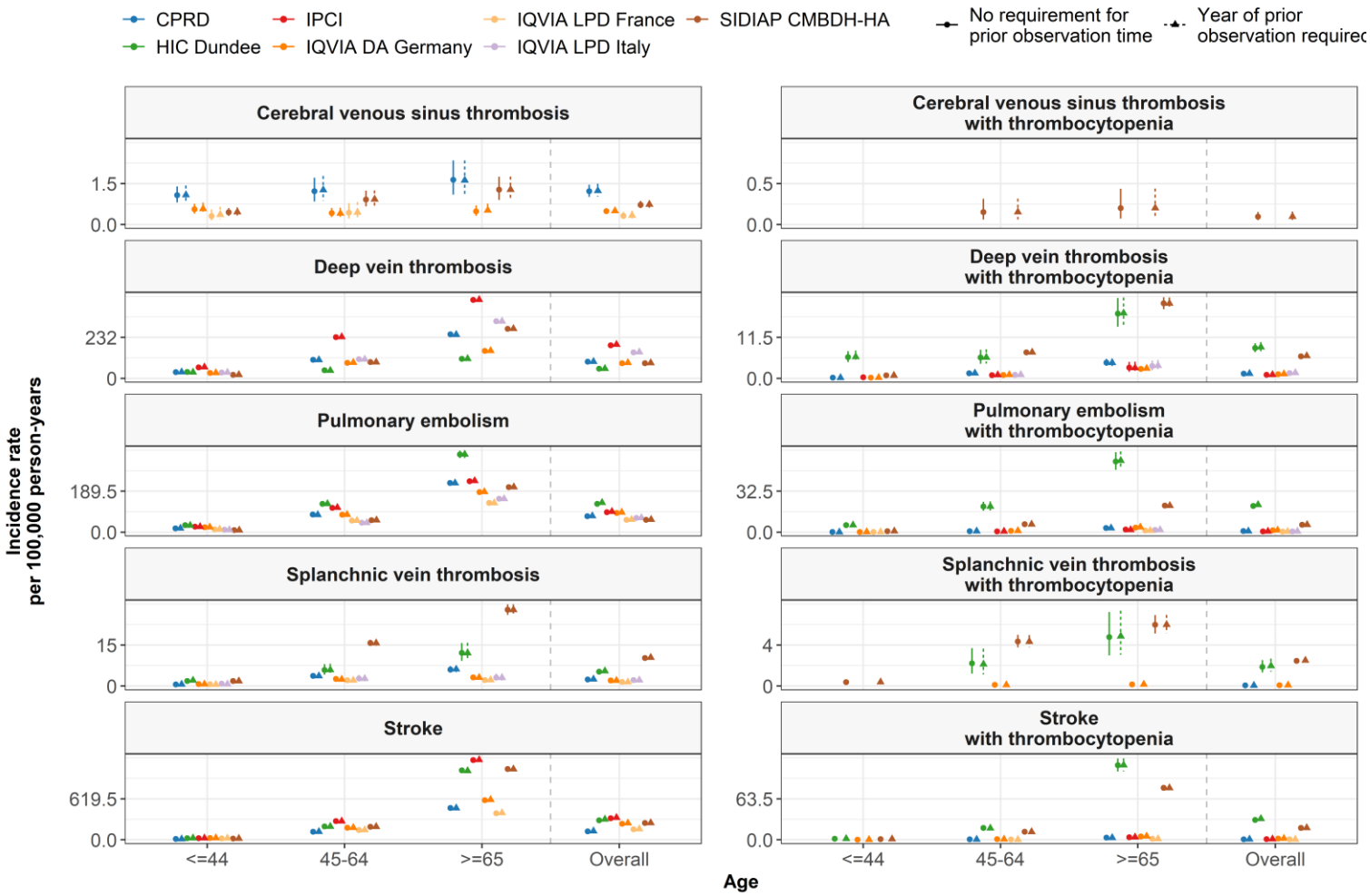
Figure 3. Comorbidities and prior medication use among patients with TTS compared to the overall study population. Any characteristic seen in less than five people in a cohort is not reported.



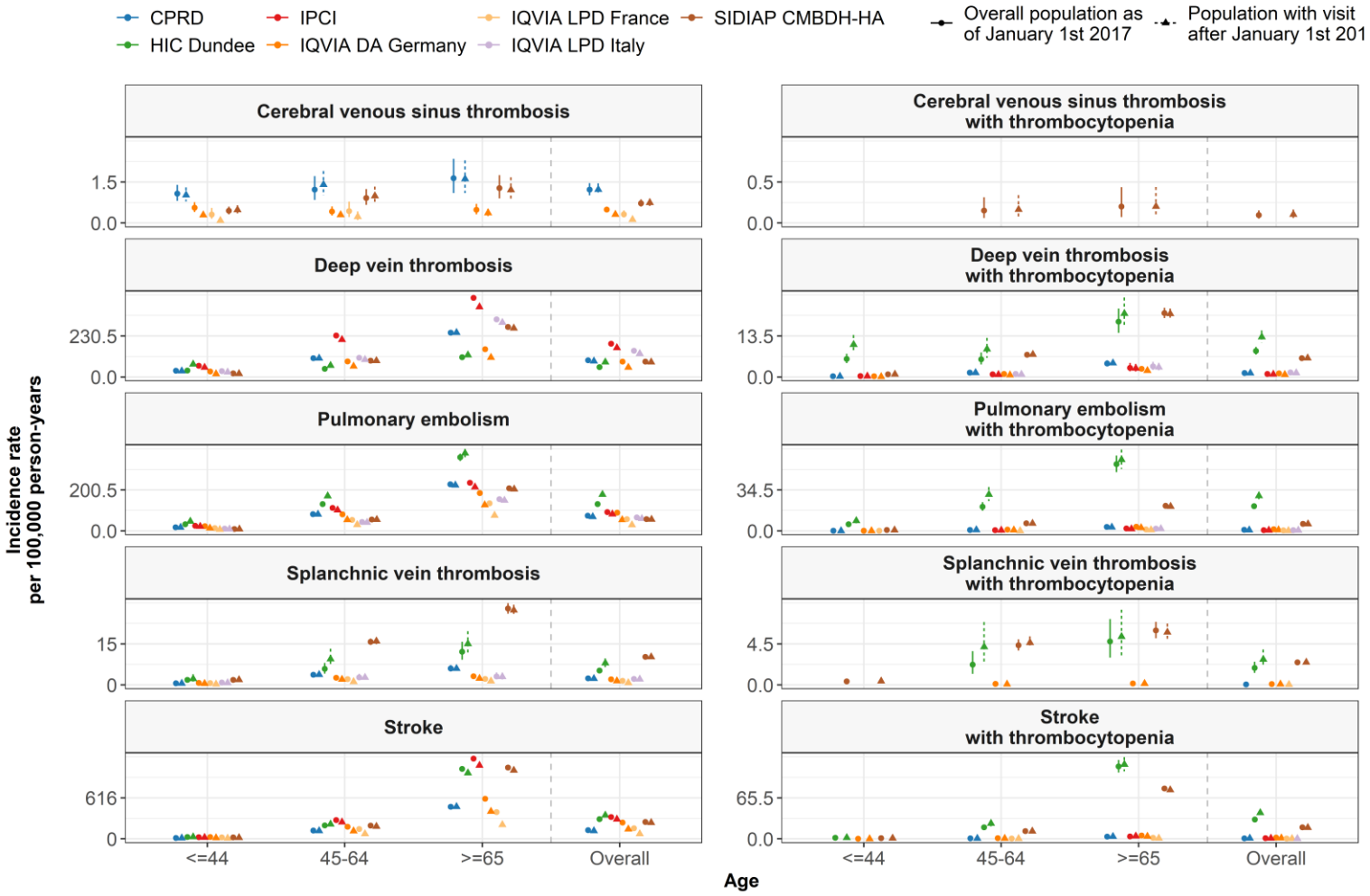
medRxiv preprint doi: https://doi.org/10.1101/2021.05.12.21257083; this version posted September 17, 2021. The copyright holder for this preprint **(which was not certified by peer review)** is the author/funder, who has granted medRxiv a license to display the preprint in perpetuity.
It is made available under a CC-BY 4.0 International license .

Appendix

A1. Impact of requiring a year of prior history for study participants to be included in the analysis



A2. Impact of defining study populations based on a visit



medRxiv preprint doi: https://doi.org/10.1101/2021.05.12.21257083; this version posted September 17, 2021. The copyright holder for this preprint (which was not certified by peer review) is the author/funder, who has granted medRxiv a license to display the preprint in perpetuity.
It is made available under a CC-BY 4.0 International license .

A3. Heterogeneity in the incidence rates across databases

In this study variation in incidence rates has been seen across the databases used. Below we discuss the reasons for this, with a particular focus on the results for deep vein thrombosis as an example.

Analytic choices

While heterogeneity between studies in the literature can often in part be explained by variation in analytic choices, this is not the case in this study where we applied the same analytic code to each database.

Differences in setting

Variation in the ´true´ background incidence rates across countries is not in itself unexpected, given differences in population demographics, the prevalence of risk factors for the events of interest, and so on. For example, risk factors for deep vein thrombosis include obesity (https://www.sciencedirect.com/science/article/pii/S0140673605718808), and the prevalence of obesity is well-known to vary substantially across countries (https://link.springer.com/article/10.1007/s00394-014-0746-4).

Differences in source coding systems

Medical vocabularies vary and the databases used in the study use Read codes (CPRD), ICPC (IPCI), ICD-9 (LPD Italy), ICD-10 (DA Germany and LPDD France), and ICD-10CM (SIDIAP) to represent condition-related concepts. These coding systems differ in the way that they describe clinical events (in particular their granularity) and this can have a meaningful impact on research findings. For example, this can be seen in the literature by the impact on research findings when databases switched from using ICD-9 to ICD-10 codes. (https://onlinelibrary.wiley.com/doi/abs/10.1002/pds.4563).

In our study, a substantial impact can be seen for some databases when using the narrow definition for DVT rather than the broad one, see Figure 1 and Figure 2 below. The most common source codes are for our narrow DVT definition are “Deep vein thrombosis” (CPRD – Read code), “Phlebitis and thrombophlebitis of other deep vessels of lower extremities” (DA Germany and LPD France – ICD-10 code), “Other venous embolism and thrombosis of inferior vena cava” (LPD Italy- ICD-9 code), and “Phlebitis and thrombophlebitis of unspecified deep vessels of unspecified lower extremity” (SIDIAP – ICD-10-CM). While broadening the definition of DVT had only modest effect on the results for

medRxiv preprint doi: https://doi.org/10.1101/2021.05.12.21257083; this version posted September 17, 2021. The copyright holder for this preprint (which was not certified by peer review) is the author/funder, who has granted medRxiv a license to display the preprint in perpetuity.
It is made available under a CC-BY 4.0 International license .

CPRD and IPCI, in did have a substantial impact for DA Germany (where the most common included code became "Phlebitis and thrombophlebitis of lower extremities, unspecified"), for LPD France (where the most common included code became "Phlebitis and thrombophlebitis of unspecified site"), for LPD Italy (where the most common included code became "Other venous embolism and thrombosis of unspecified site"), and SIDIAP (where the most common included code became "Phlebitis and thrombophlebitis of unspecified deep vessels of unspecified lower extremity"). This provides one example of the underlying variation between the databases in how the presentation of a particular is captured in the medical record. While using a common data model facilitates the analysis of the data sources in a standardised manner, this underlying heterogeneity in the source data is still reflected to an important degree.

Figure A3.1. Estimated incidence rates for DVT using a narrow definition

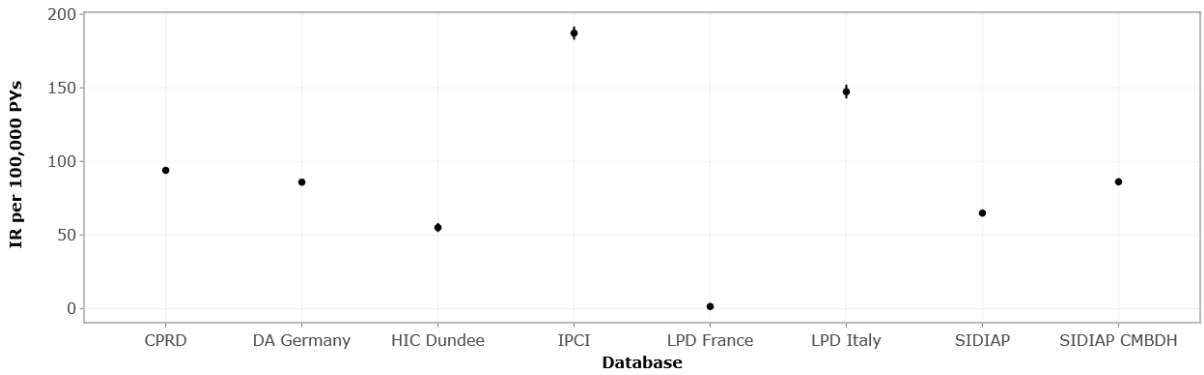
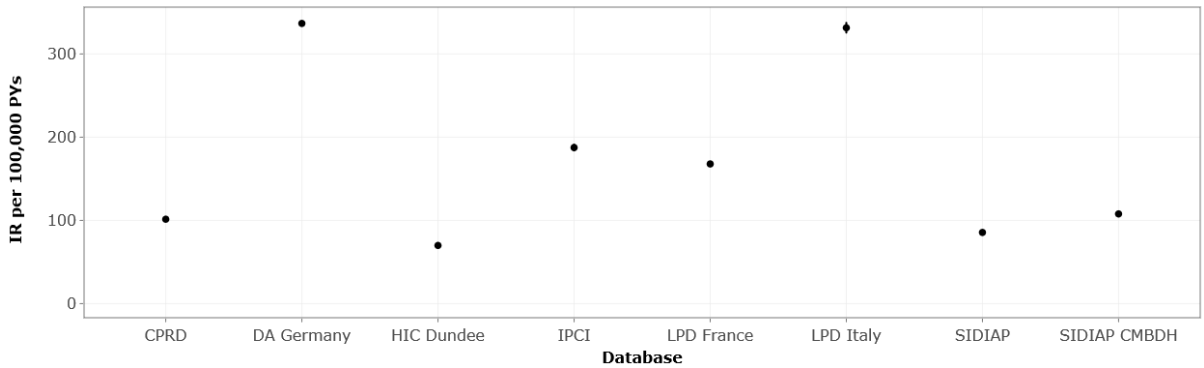


Figure A3.2. Estimated incidence rates for DVT using a broad definition



medRxiv preprint doi: https://doi.org/10.1101/2021.05.12.21257083; this version posted September 17, 2021. The copyright holder for this preprint (which was not certified by peer review) is the author/funder, who has granted medRxiv a license to display the preprint in perpetuity.
It is made available under a CC-BY 4.0 International license .

Differences in capture of hospital events

Many of the events considered in our study can be expected to involve hospitalisation. Consequently, the degree to which diagnoses made in the hospital setting are reflected in the data sources used can be expected to have a large effect on the capture of events. We see this for SIDIAP where we ran the analysis in both the database with only primary care records and again with linkage at a patient-level to hospital records, SIDIAP CMBDH-HA.

An individual in the community with DVT may require hospital treatment and DVT may also occur in persons hospitalised for other reasons. For our narrow definition, we see that the incidence rate of DVT increases from 64.9 per 100,000 person-years in SIDIAP to 86.2 per 100,000 person-years in SIDIAP CMBDH-HA, see figure 3.

While we can expect patient-level linkage of the other data sources to hospital level data to impact their results, it will not necessarily be to the same degree as in SIDIAP. In Catalonia, general practitioners are generally able to view patient hospital records during a consultation in a separate program to that used to view primary care records, with only the latter is captured in the SIDIAP database. Consequently, for patient care there is not a need to copy across what is seen in the system with the hospital record into the system with the primary care record which is captured by SIDIAP. This is not the case in the UK, however, where there is a need to add the information from letters received by general practitioners from secondary care facilities into the electronic medical record. Consequently, the impact of hospital linkage is likely to be context-dependent, as well as varying depending on the event under consideration.

medRxiv preprint doi: https://doi.org/10.1101/2021.05.12.21257083; this version posted September 17, 2021. The copyright holder for this preprint **(which was not certified by peer review)** is the author/funder, who has granted medRxiv a license to display the preprint in perpetuity.
It is made available under a CC-BY 4.0 International license .

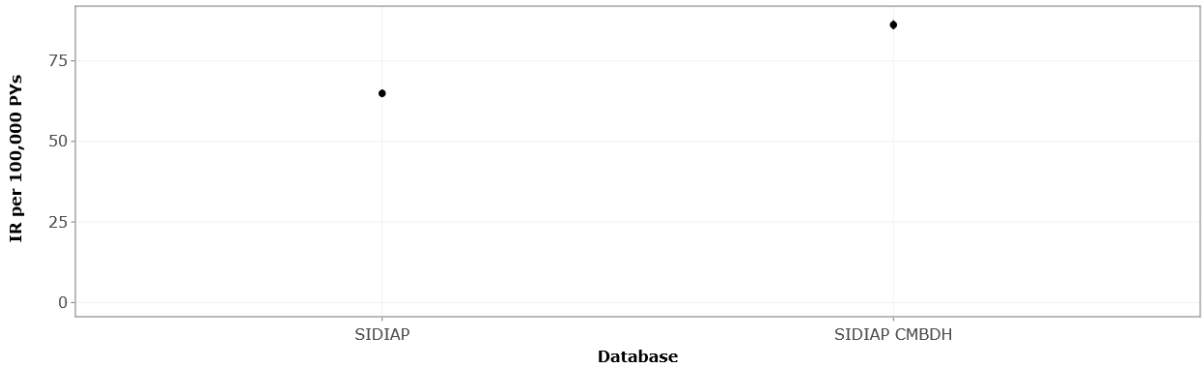



Figure A3.3. Estimated incidence rates for DVT using a narrow definition in SIDIAP and SIDIAP CMBDH-HA